

# ***WINNER ENGINEERING INTERNAL TRAINING SERIES***

## MODULE II

### Basic ACMV Knowledge & Design

# ***CONTENT***

1. Ducting Distribution System
2. Piping System
3. Air Terminals
4. Water Fittings
5. Air Side Equipment
6. Water Side Equipment

# 1.1 DUCT DISTRIBUTION SYSTEM

Purpose is to distribute the air from the source to the final destination point.

Things to consider:

## Materials

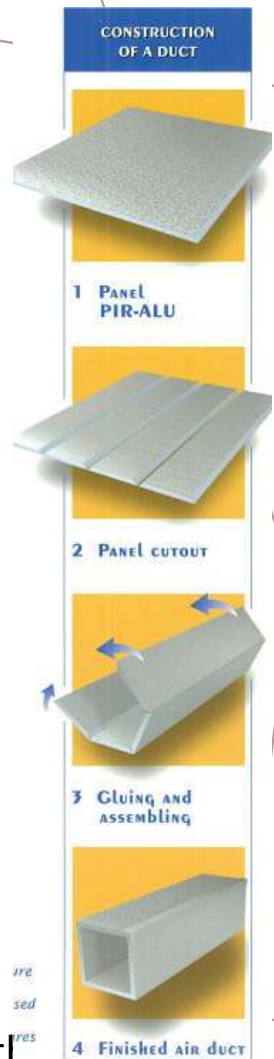
- Galvanised steel (most common usage)
- Stainless steel (kitchen duct, smoke extraction system)
- Pre-insulated duct (can be proposed due to certain advantages, eg; lightweight, time saving for installation)

## Shape for steel duct

- Round duct (cost saving in terms of constructability, material, and installation)
- Square duct, more flexible for height and width ratio for site constraints.

## Thickness

- Duct sheet thickness varies accordingly to the required air flow and static pressure in the duct (Refer to installation details table)



## TYPE OF DUCTING SYSTEM

1. SAD – Supply Air Duct (Cold air from equipment to air terminal)
2. FAD – Fresh Air Duct (Outdoor air intake to equipment)
3. RAD – Return Air Duct ( Recirculated air from the room to the equipment)
4. EAD – Exhaust Air Duct ( Extracted air from indoor to outdoor)
5. PAD – Pre-cool Air Duct ( Pre-cool air from equipment to equipment or air terminal)

## INSULATION THICKNESS FOR DUCT IN DIFFERENT CONDITIONS

| TYPE                    | CONDITION          | INSULATION THICKNESS (MM) |
|-------------------------|--------------------|---------------------------|
| SUPPLY                  | NO AC SPACE        | 50                        |
| SUPPLY, FRESH AIR       | AC SPACE           | 25                        |
| RETURN, EXHAUST         | NO AC SPACE        | 25                        |
| SUPPLY,RETURN,EXHAUST   | EXPOSED TO WEATHER | 50 / HYBRID PLASTER       |
| SUPPLY, RETURN, EXHAUST | BELOW THE ROOF     | 50                        |

## **1.2 DUCT SIZING**

Proper duct size is important to ensure the stability of the air distribution system.

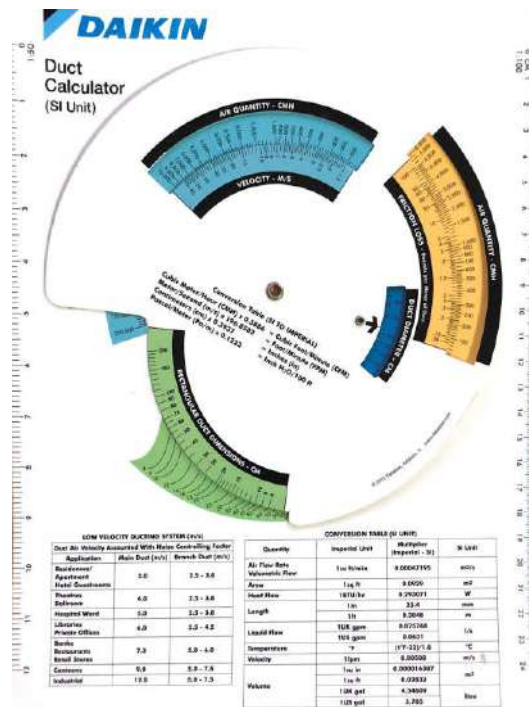
- Oversized duct resulted in material wastage, low air flow, face velocity and throwing distance from air terminal
- Undersized duct resulted in high static pressure and unwanted noise generation due to turbulence and friction.

### **Precon to duct design and sizing.**

1. Understand the project scope of work, find out with OIC to check with Main Contractor if PID or round duct is acceptable.
2. Round duct is preferred for cost saving.

### **Duct sizing by using**

1. Ductulator hardcopy
2. Ductulator Software



| Duct Sizing |             |          |      |           |        |                            |     |       |          |                   |                    |                       |                     |                 |                                                               |       |    |
|-------------|-------------|----------|------|-----------|--------|----------------------------|-----|-------|----------|-------------------|--------------------|-----------------------|---------------------|-----------------|---------------------------------------------------------------|-------|----|
| Block       | Description | Key in   |      | Key in    |        | Calculated                 |     |       |          |                   |                    |                       |                     |                 |                                                               | Check |    |
|             |             | Air Flow |      | Duct Size |        | Aspect Ratio<br>(c-t to b) | De  | Area  | Velocity | Velocity Pressure | Hydraulic Diameter | Absolute<br>Roughness | Pneumatic<br>Number | Friction Factor | Duct Pressure Loss (calculated)<br>(Our Standard 0.8 to 1 ps) |       |    |
|             |             | CMH      | L/S  | W (m)     | D (mm) |                            |     |       |          |                   |                    |                       |                     |                 |                                                               | mm    | m² |
|             | FAF SL-1    | 11000    | 3056 | 700       | 630    | 1.1                        | 738 | 0.428 | 7.1      | 30.75             | 678                | 0.05                  | 319077.363          | 0.015           | 0.70                                                          |       |    |
|             |             | 5500     | 1528 | 700       | 400    | 1.8                        | 573 | 0.258 | 5.9      | 21.14             | 509                | 0.05                  | 790173.5416         | 0.016           | 0.70                                                          |       |    |
|             |             | 265      | 100  | 500       | 250    | 2.0                        | 456 | 0.163 | 0.6      | 8.23              | 422                | 0.05                  | 16750.11742         | 0.028           | 0.82                                                          |       |    |
|             |             | 720      | 200  | 500       | 250    | 2.0                        | 456 | 0.163 | 1.2      | 0.91              | 422                | 0.05                  | 33500.23405         | 0.024           | 0.86                                                          |       |    |
|             |             | 1080     | 300  | 500       | 250    | 2.0                        | 456 | 0.163 | 1.8      | 2.04              | 422                | 0.05                  | 50750.35277         | 0.022           | 0.81                                                          |       |    |
|             |             | 1440     | 400  | 500       | 250    | 2.0                        | 456 | 0.163 | 2.5      | 3.62              | 422                | 0.05                  | 67900.4697          | 0.021           | 0.80                                                          |       |    |
|             |             | 1800     | 500  | 500       | 250    | 2.0                        | 456 | 0.163 | 3.1      | 5.65              | 422                | 0.05                  | 93750.58712         | 0.020           | 0.77                                                          |       |    |
|             |             | 2160     | 600  | 500       | 250    | 2.0                        | 456 | 0.163 | 3.7      | 8.14              | 422                | 0.05                  | 120500.7045         | 0.019           | 0.77                                                          |       |    |
|             |             | 2520     | 700  | 500       | 250    | 2.0                        | 456 | 0.163 | 4.3      | 11.08             | 422                | 0.05                  | 147300.822          | 0.018           | 0.74                                                          |       |    |
|             |             | 2880     | 800  | 500       | 250    | 2.0                        | 456 | 0.163 | 4.9      | 14.67             | 422                | 0.05                  | 174000.9394         | 0.018           | 0.74                                                          |       |    |
|             |             | 3240     | 900  | 700       | 560    | 1.4                        | 645 | 0.327 | 5.5      | 18.35             | 583                | 0.05                  | 213814.5084         | 0.016           | 0.82                                                          |       |    |
|             |             | 3600     | 1000 | 80        | 250    | 1.6                        | 210 | 0.035 | 0.6      | 3.49              | 288                | 0.05                  | 29949.52496         | 0.026           | 0.87                                                          |       |    |
|             |             | 3960     | 1100 | 700       | 560    | 1.4                        | 645 | 0.327 | 6.1      | 21.88             | 583                | 0.05                  | 30155.88167         | 0.020           | 0.80                                                          |       |    |
|             |             | 4320     | 1200 | 700       | 280    | 2.5                        | 491 | 0.189 | 5.6      | 24.60             | 420                | 0.05                  | 39059.19717         | 0.019           | 0.84                                                          |       |    |
|             |             | 4680     | 1300 | 800       | 280    | 2.9                        | 420 | 0.180 | 5.6      | 14                | 378                | 0.05                  | 40580.80791         | 0.023           | 0.84                                                          |       |    |
|             |             | 5040     | 1400 | 700       | 250    | 2.8                        | 534 | 0.224 | 4.9      | 24.25             | 467                | 0.05                  | 150732.1756         | 0.017           | 0.80                                                          |       |    |
|             |             | 5400     | 1500 | 800       | 250    | 2.4                        | 434 | 0.135 | 4.8      | 14.06             | 353                | 0.05                  | 113217.4649         | 0.019           | 0.74                                                          |       |    |
|             |             | 5760     | 1600 | 400       | 250    | 1.6                        | 344 | 0.093 | 2.3      | 3.28              | 308                | 0.05                  | 47653.00346         | 0.022           | 0.84                                                          |       |    |
|             |             | 6120     | 1700 | 500       | 250    | 2.0                        | 381 | 0.114 | 5.0      | 9.16              | 323                | 0.05                  | 86376.61233         | 0.020           | 0.84                                                          |       |    |
|             |             | 6480     | 1800 | 600       | 400    | 1.5                        | 533 | 0.223 | 5.9      | 20.64             | 400                | 0.05                  | 106386.6636         | 0.017           | 0.79                                                          |       |    |
|             |             | 6840     | 1900 | 400       | 300    | 1.3                        | 378 | 0.112 | 0.8      | 0.43              | 343                | 0.05                  | 29169.21997         | 0.027           | 0.80                                                          |       |    |
|             |             | 7200     | 2000 | 300       | 400    | 0.8                        | 378 | 0.112 | 2.5      | 3.7               | 343                | 0.05                  | 56880.05872         | 0.021           | 0.82                                                          |       |    |
|             |             | 7560     | 2100 | 200       | 200    | 1.0                        | 219 | 0.030 | 27.4     | 481.05            | 250                | 0.05                  | 363506.0115         | 0.017           | 0.79                                                          |       |    |
|             |             | 7920     | 2200 | 150       | 150    | 1.0                        | 164 | 0.021 | 24.4     | 368.57            | 150                | 0.05                  | 243076.7375         | 0.019           | 0.61                                                          |       |    |
|             |             | 8280     | 2300 | 400       | 280    | 1.3                        | 378 | 0.112 | 1.4      | 1.12              | 343                | 0.05                  | 11009.8523          | 0.024           | 0.80                                                          |       |    |
|             |             | 8640     | 2400 | 300       | 280    | 1.1                        | 267 | 0.066 | 1.0      | 1.24              | 240                | 0.05                  | 47460.07140         | 0.021           | 0.80                                                          |       |    |
|             |             | 9000     | 2500 | 200       | 210    | 0.9                        | 210 | 0.015 | 2.4      | 1.49              | 180                | 0.05                  | 3996.52496          | 0.026           | 0.87                                                          |       |    |
|             |             | 9360     | 2600 | 300       | 200    | 1.5                        | 267 | 0.056 | 7.3      | 22.06             | 240                | 0.05                  | 116479.104          | 0.019           | 0.76                                                          |       |    |
|             |             | 9720     | 2700 | 500       | 250    | 2.0                        | 381 | 0.114 | 7.2      | 30.93             | 333                | 0.05                  | 158625.1102         | 0.018           | 0.76                                                          |       |    |
|             |             | 10080    | 2800 | 600       | 400    | 1.5                        | 533 | 0.223 | 6.3      | 22.43             | 400                | 0.05                  | 194536.8217         | 0.017           | 0.76                                                          |       |    |
|             |             | 10440    | 2900 | 500       | 400    | 1.3                        | 489 | 0.180 | 5.5      | 18.15             | 488                | 0.05                  | 162026.3609         | 0.017           | 0.71                                                          |       |    |

## 1.3 DUCT SIZING RULE OF THUMB

### Considerations

Static pressure of system, air flow, velocity, noise.

#### **Approach 1**

- Sizing based on static loss, (standard 0.6 to 1.0)Pa/m
  - indicates the amount of static losses in the system per meter duct run.
  - good practice to size main duct at 0.6 for lesser static loss.

#### **Approach 2**

- Sizing based on velocity
  - If the duct sized based on approach 1 has a **velocity of >10m/s**, to size again based on max. 10m/s.
  - Considered as secondary approach due to spaces like library which require lesser dBA ambient.

$$\text{*Velocity (m/s) = Air Flow (m}^3\text{/s) / Duct Area (m}^2\text{)}$$

## **2.1 PIPING SYSTEM**

- To delivery as a pathway for the flow of medium (water/refrigerant) of heat transfer to occur.

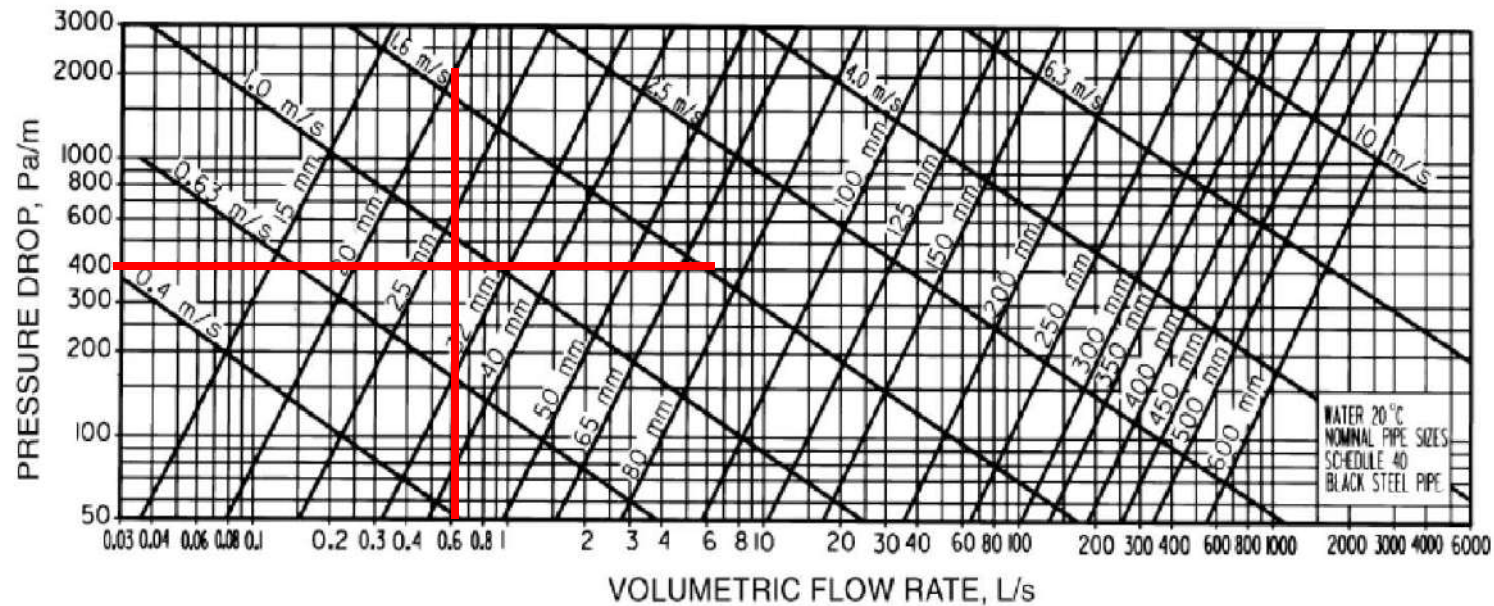
### **Types of Common Piping System**

1. CHWS - Chilled Water Supply
2. CHWR - Chilled Water Return
3. CDP - Condensate Drain Pipe
4. REF. - Refrigerant pipe
5. \*CHW Loop - Chilled water loop from chiller evaporator to air side equipment
6. \*CW Loop - Condenser water loop from chiller condenser to cooling tower.

## 2.2 PIPE SIZING

### Chilled Water Piping System

**Table 1:** Friction Loss for Water in Commercial Steel Pipe (Scheduled 40)



## **2.2 PIPE SIZING**

### **CONDENSATE DRAIN PIPE**

| COOLING CAPACITY (KW) | DRAIN PIPE SIZE (MM) |
|-----------------------|----------------------|
| BELOW 17              | ø25                  |
| 17.1 ~ 35             | ø32                  |
| 35.1 ~ 175            | ø40                  |
| ABOVE 175             | ø50                  |

## **3.1 AIR TERMINAL**

- To dissipate or extract air from the ductwork distribution system

### **Types of Common Air Terminal**

1. SAG,SAL : Supply Air Grille, Supply Air Louvre
2. RAG : Return Air Grille
3. EAG,EAL : Exhaust Air Grille, Exhaust Air Louvre
4. LSAD,LRAD : Linear Supply Air Diffuser, Linear Return Air Diffuser

### **Types of Features**

- Louvre : Insect/vermin proof, weather proof
- Grille : Anti-condensation, SDV,DDV,2-way,3-way,4-way
- Linear Diffuser : 2 slot,4 slot, (Depends on air flow and throwing distance requirement)

## **3.2 SIZING AIR TERMINAL RULE OF THUMB**

- Undersized air terminal resulted in high static pressure loss, and noise generation
- Sizing is normally done based on the following common practice parameters
  1. Face velocity of maximum 2.5m/s
  2. Free area opening of 50% and 70% for louvres and grilles respectively

### **Approach 1**

- Manual calculation

### **Approach 2**

- Excel sheet (refer to e-cabinet)

$$\text{Velocity} = \text{Air Flow} \div \text{Area}$$
$$\text{m}^3/\text{s} = (\text{m}^3/\text{s}) \div \text{m}^2$$

## 3.2 SIZING AIR TERMINAL RULE OF THUMB

### Approach 1

Manual calculation principle

Velocity = Air Flow  $\div$  Area

Area = Air Flow  $\div$  Velocity

$m^2 = (m^3/s) \div m/s$

$m^2 = (CMH \div 3600) \div m/s$

For external louvre of 50% opening, and face velocity of 2.5m/s,

Therefore,  $0.5m^2 = (CMH \div 3600) \div 2.5m/s$

\*By limiting the face velocity to 2.5m/s, the static pressure loss across the louvres would be limited to 50Pa or less.

## **3.2 SIZING AIR TERMINAL RULE OF THUMB**

### **Approach 2**

Excel Sheet

